

QoS Aware Virtual Network Embedding in Space-Air-Ground-Ocean Integrated Network

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Abstract—The space-air-ground-ocean integrated network (SAGOI-Net) has become the focus of research in recent years, which has the characteristics of wide coverage and strong adaptability. However, due to the influence of multiple heterogeneous network segments, this network is unable to provide excellent quality of service (QoS). Based on the software-defined network and virtual network architecture, we abstract SAGOI-Net as a three-layer heterogeneous physical network resource, and propose a multi-domain virtual network embedding solution to optimize QoS. Specifically, before virtual network embedding, we collected SAGOI-Net's resource information through software-defined network and modeled it. In the virtual network embedding process, we first classify the virtual network request through K-means, and dynamically adjust the reward function to use reinforcement learning to solve the optimal virtual network embedding strategy. Finally, simulation experiments verify the effectiveness of the scheme.

Index Terms—Space-air-ground-ocean Integrated Network, Software Defined Network, Virtual Network Architecture, Virtual Network Embedding, Reinforcement Learning.

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I. INTRODUCTION

IN RECENT years, the emergence of a variety of new IoT applications has brought huge challenges to traditional ground communication networks [1], [2]. For example, the development of intelligent transportation systems (ITS) has put forward higher requirements for the service quality of vehicle communication networks [3], [4]. However, traditional vehicle communication networks rely on terrestrial communications networks to provide services [5]. Due to factors such as deployment, capacity, and response delays, terrestrial communications networks gradually cannot meet the needs of the next generation of intelligent transportation systems [6], [7]. Therefore, the space-air-ground-ocean integrated network (SAGOI-Net) is introduced to provide a multi-dimensional vehicle communication network with comprehensive coverage and low response delay for vehicles traveling around the world [8].

SAGOI-Net is a new type of hierarchical wireless network structure, which is composed of multiple heterogeneous networks, namely space network, air network, ground network and ocean network [9]. These heterogeneous networks can independently improve services to ITS [10], [11], and can also provide services through mutual operation. As revealed in Fig. 1, in military operations, the cooperation of satellite networks, air networks, ground networks, and ocean networks is needed to complete the high quality of service (QoS) requirements.

However, as a multi-tier network architecture, SAGOI-Net is affected by multiple heterogeneous network segments during operation, and cannot efficiently use network resources [12], [13], so it is difficult to provide excellent quality of service QoS [14]. How to overcome the complexity of the network and efficiently allocate network resources to meet the differentiated QoS requirements of users has become one of the challenges of SAGOI-Net. To address this problem, we abstract SAGOI-Net as a three-layer heterogeneous network and propose a multi-domain virtual network embedding algorithm, which gives a solution in the perspective of network virtualization. And based on this, we use K-means clustering technique to sense the QoS categories of VNRs to meet the differentiated QoS requirements of users. Specifically, the contributions of this paper are as follows:

- 1) First, we design a multilayer policy network as a reinforcement learning agent for virtual network embedding. This policy network perceives the topological properties of the network to extract feature matrices to derive the

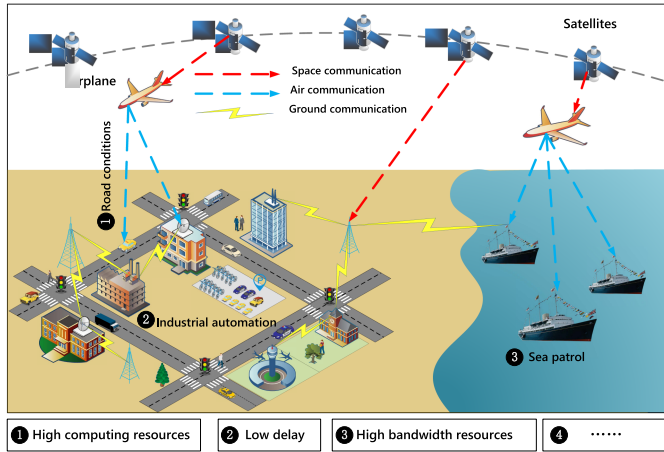


Fig. 1. Differentiated QoS application scenarios in SAGOI-Net.

embedding scheme of virtual nodes, which can achieve efficient allocation of network resources.

- 2) Second, we use the K-means clustering technique to sense the QoS categories of virtual network requests (VNRs) and dynamically adjust the reward function used by the agent to meet the differentiated QoS requirements based on the results. This can make the algorithm more adaptable to SAGOI-Net that covers complex application scenarios.
- 3) We designed several sets of comparison experiments to verify the performance of the algorithm. The simulation results show that our algorithm has excellent performance in terms of delay, acceptance rate, and revenue.

The remainder of this article is arranged as follows. Section II firstly introduces the application of software-defined networking (SDN) under SAGOI-Net and the application of virtual architecture in SAGOI-Net, and then analyzes the current research status. Section III establishes the topological model of the substrate network and VNR. Section IV respectively describes the multi-domain virtual network embedding algorithm for QoS optimization. In the Section V, the experimental results of this scheme are analyzed. Section VI summarizes the article and prospects future work.

II. RELATED WORK

A. Application of SDN Architecture Under SAGOI-Net

As a new type of network architecture, SDN has excellent flexibility and reconfigurability [15], [16]. At present, SDN has been widely used in ground network systems. In order to maximize the role of SDN in the integration of satellite networks, air networks and ground networks, many scholars have proposed many SDN-based satellite ground fusion or satellite air fusion network architecture by deploying controllers on satellites, ground and other locations [17], [18], [19], [20].

Literature [17] designed a multi-layer satellite network architecture based on SDN. The authors deployed the controller on geosynchronous orbit (GEO) satellites and used the extensive coverage of GEO satellites to control medium earth orbit

(MEO) and low earth orbit (LEO) satellites. In [18], the author Yang et al. deployed the controller in the ground network and proposed a network architecture based on SDN that integrates satellite and ground. This architecture ensures seamless hand-over between LEO satellites, and at the same time provides flexible traffic management and fine-grained QoS for satellite and ground converged networks. Literature [19] proposed a network architecture called SERV-LCE, which is also based on SDN implementation. SERV-LCE deploys controllers in GEO satellites, ground satellite gateways and ground data centers to collect relevant information from satellites and ground network forwarding planes. In [20], the author Zhang et al. designed an SDN-based SAGOI-Net architecture for vehicle networks. In the article, they suggested that SDN controllers should be deployed in a hierarchical architecture to control three heterogeneous network segments: satellite, air, and ground. The architecture uses a centralized control mode, and all controllers are deployed on the ground.

The above-mentioned SDN-based architecture mainly deploys the controller on the ground to control the satellite or aerial platform, which causes the controller to be unable to receive real-time status information from the satellite and aerial platform, and the communication delay is also high. In addition, when user demand increases, the frequent switching of satellites and aerial platforms also puts forward higher demands on link bandwidth. In order to solve the above problems, the author of [21] optimized the current SDN-based SAGOI-Net architecture.

B. Research Status of SAGOI-Net Based on Virtual Network Architecture

Because the virtual network architecture has made breakthroughs in the research of traditional ground communication networks, some scholars have proposed that the virtual network architecture be used in SAGOI-Net. The virtual network architecture has significant advantages in dealing with differentiated QoS for different applications. As excellent representatives of network virtualization, SDN and network function virtualization are considered to be enabling technologies for the flexible and effective integration of heterogeneous networks, and can provide innovative solutions for the orchestration of heterogeneous network resources [22].

In order to optimize the load balancing of network communication, reference [23] studied a software-defined SAGOI-Net routing algorithm. Based on the characteristics of the SDN model and the dynamic changes of SAGOI-Net's topology, the authors considered the multi-dimensionality of resources and energy consumption, which effectively reduced the end-to-end delay and packet loss rate. Literature [24] proposed a SAGOI-Net reconfigurable service framework based on service function chain. The framework modeled the realization of service function chain and virtual network functions as integer nonlinear programming problems. The authors proposed a heuristic greedy algorithm to balance the resource consumption of different network nodes. The results proved that this algorithm can improve resource utilization efficiency. Du et al. [25] studied spectrum sharing and interference control technology based on SDN. They

proposed a spectrum sharing and service offloading mechanism to realize the cooperative relationship between the ground BSs and the beam group of the satellite ground communication system. In this mode, the communication between satellites and the ground effectively realized frequency sharing and traffic offloading.

C. Research Status Analysis

Through the analysis of the above-mentioned algorithm and the virtual network embedding research under the SAGOI-Net, we found that they have the following problems.

- 1) In the past, most virtual network embedding algorithms used heuristic algorithms to solve problems, which could easily fall into local optimal solutions.
- 2) The current research on multi-domain virtual network embedding algorithms mostly focuses on ground networks, and there is almost no research on heterogeneous, hierarchical multi-domain networks.
- 3) The existing researches only focus on the impact of SAGOI-Net's heterogeneity and time-variability on users' QoS requirements, and have not specifically modeled the physical network resources of the three network segments. Therefore, it is difficult to achieve efficient scheduling of resources and cannot meet the differentiated QoS requirements of users. Therefore, in view of the above problems, after considering the heterogeneity and time-varying nature of SAGOI-Net, we model the SAGOI-Net hierarchically based on the SDN network architecture based on the literature [21]. Then we propose a reinforcement learning (RL)-assisted QoS-aware multi-domain virtual network embedding algorithm. In view of the influence of the time-varying nature of SAGOI-Net's topology on virtual network embedding, we have also re-embedded the VNR subgraphs with embedding failures to ensure stable QoS.

III. MODEL BUILDING AND PROBLEM DESCRIPTION

A. Problem Description

For privacy and confidentiality considerations, infrastructure provider (InP) often does not disclose all physical network information to SDN, which is also one of the problems faced by traditional multi-domain virtual network embedding algorithms. In addition, if the InP uploads all the information of the physical network to the SDN, a large amount of transmission signaling overhead and calculation overhead will be generated. Therefore, in order to solve this problem, before virtual network embedding, we stipulated that each InP only needs to upload the following basic information: the network topology of each InP at different times, the CPU, bandwidth and delay of each node and intra-domain in the network, bandwidth and delay information of inter-domain links.

Moreover, due to the time-varying nature of SAGOI-Net, the network nodes in SAGOI-Net are moving. For example, a ship is receiving communication services provided by satellite A, and due to the movement of satellite A, the ship sails out of the

coverage area of satellite A. At this time, SAGOI-Net's topology has changed. We use formula 1 to adapt to the uncertainty caused by topology changes. We first divide the network topology at time T into consecutive $n \Delta t$ and assume that the VNR arrives at the moment t . The number of VNRs changes with time. Then, the probability of a node working properly is calculated based on the number of network requests carried by the SAGOI-Net at moment t , as follows.

$$[e^{-0.004(m-1)}, e^{-0.004m}], \quad (1)$$

where m denotes the number of VNRs that have been carried in SAGOI-Net.

B. Substrate Network

We model the topological snapshot of the substrate network at time t as a weighted undirected graph $G^S = \{N^S, L^S, A^S\}$, where N^S represents the network node collection of SAGOI-Net, L^S is the network link collection of SAGOI-Net, and A^S represents the attribute collection of SAGOI-Net. Network node collection $N^S = \{N_S^S, N_A^S, N_{G\&O}^S\}$, among them, N_S^S is a set of satellite network nodes, N_A^S is a set of air network nodes, and $N_{G\&O}^S$ is a set of ground-ocean network nodes. Network link collection $L^S = \{L_S^S, L_A^S, L_{G\&O}^S, L_{S,A}^S, L_{A,G\&O}^S, L_{S,G\&O}^S\}$, where L_S^S , L_A^S and $L_{G\&O}^S$ respectively represent the link set of the satellite network, the link set of the air network and the link set of the ground-ocean network. $L_{S,A}^S$ is the set of inter-domain links between satellite network nodes and air network nodes. $L_{A,G\&O}^S$ is the set of inter-domain links between air network nodes and ground-ocean network nodes. $L_{S,G\&O}^S$ is the set of inter-domain links between satellite network nodes and ground-ocean network nodes. Network attribute collection $A^S = \{CPU_{N_S^S}, CPU_{N_A^S}, CPU_{N_{G\&O}^S}, BW_{L_S^S}, BW_{L_A^S}, BW_{L_{G\&O}^S}, D_{L_S^S}, D_{L_A^S}, D_{L_{G\&O}^S}\}$, among them, $CPU_{N_S^S}$ represents the computing resource of the satellite network node, $CPU_{N_A^S}$ represents the computing resource of the air network node, and $CPU_{N_{G\&O}^S}$ represents the computing resource of the ground-ocean network node. $BW_{L_S^S}$, $BW_{L_A^S}$ and $BW_{L_{G\&O}^S}$ respectively represent the bandwidth of satellite network link, air network link, and ground-ocean network link. $D_{L_S^S}$, $D_{L_A^S}$ and $D_{L_{G\&O}^S}$ respectively represent the delay of satellite network link, air network link, and ground-ocean network link.

C. Virtual Network Request

Similar to the underlying physical network, VNRs are modeled as weighted undirected graph $G^V = \{N^V, L^V, A^V\}$, where N^V denotes the virtual nodes set, L^V indicates the virtual links set, and $A^V = \{CPU_{N^V}, CD_{N^V}, BW_{L^V}, D_{L^V}\}$ is the attribute set of virtual network requests. In expression A^V , CPU_{N^V} reveals the computing resources required by the virtual nodes, CD_{N^V} is the calculation delays required by the virtual nodes, BW_{L^V} is the bandwidth resources required by the virtual links, and D_{L^V} indicates the delay requirement of the virtual links.

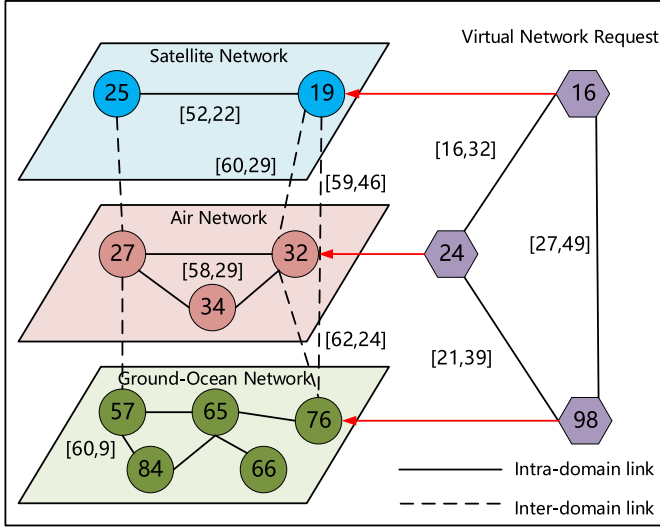


Fig. 2. Cross-domain virtual network embedding architecture. On the left is the hierarchical SAGOI-Net, and on the right is the VNR.

Fig. 2 reveals the embedding process of a multi-domain virtual network under SAGOI-Net. In the physical network, the value in the box denotes the number of computing resources of the physical node, and the value on the link indicates the number of bandwidth resources and the delay value respectively. In the virtual network request, the value in the box is the computing resource requirement of the virtual node, and the value on the link reveals the bandwidth resource requirement and the maximum delay value respectively.

D. Constraints

In order to realize multi-domain virtual network embedding, nodes and links should also meet the following constraints.

$$CPU_{n^s} \geq CPU_{n^v}, \text{ if } n^v \uparrow n^s, \quad (2)$$

$$BW_{(n_i^s, n_j^s)} \geq BW_{(n_m^v, n_n^v)}, \text{ if } (n_i^s, n_j^s) \uparrow (n_m^v, n_n^v), \quad (3)$$

$$D_{(n_i^s, n_j^s)} \leq D_{(n_m^v, n_n^v)}, \text{ if } (n_i^s, n_j^s) \uparrow (n_m^v, n_n^v), \quad (4)$$

$$Num_{max}(n^v) = 1. \quad (5)$$

In the above constraints, (2) indicates that the computing resources of physical nodes should satisfy the requirements of virtual nodes. Where $\uparrow$ denotes the mapping action, for example, $n^v \uparrow n^s$ denotes that node n^v is embedded on node n^s . Equation (3) and (4) indicate that if the virtual link (n_m^v, n_n^v) is mapped to the physical link (n_i^s, n_j^s) , the number of bandwidth resources and the delay value of the physical link (n_i^s, n_j^s) should meet the requirements of virtual links (n_m^v, n_n^v) . It is worth noting that the transmission delays of links in different network segments in SAGOI-Net are different, so we stipulate that the delay requirements of virtual links should not be less than the delay value of physical links. Equation (5) indicates that a virtual node can only be embedded on one physical node, i.e., the number of physical nodes that can be finally selected is 1.

E. Performance Indicators

We measure the performance of the algorithm in terms of revenue, cost, and acceptance rate. First, we define the revenue obtained by completing a virtual network request as follows.

$$R(G^v, t) = t_d \left[\sum_{n_i^v \in N^V} CPU_{n_i^v} + \sum_{(n_i^v, n_j^v) \in L^V} BW_{(n_i^v, n_j^v)} \right], \quad (6)$$

where t_d denotes the duration of the VNR. Correspondingly, the cost required to complete the virtual network request is calculated using the following equation.

$$Cost = t_d \left[\sum_{n_i^v \in N^V} CPU_{n_i^v} + \sum_{(n_i^v, n_j^v) \in L^V} BW_{(n_i^v, n_j^v)} hop_{(n_i^v, n_j^v)} \right], \quad (7)$$

where $hop_{(n_i^v, n_j^v)}$ denotes the number of hops of link (n_i^v, n_j^v) . From the formula, it can be seen that the revenue and cost from a virtual network request are closely related to the amount of resources it requires. Subsequently, the long-term average revenue and the long-term average revenue-cost ratio are defined by (8) and (9).

$$LAR = \lim_{T \rightarrow \infty} \frac{\sum_t R(G^v, t)}{T}. \quad (8)$$

$$LRC = \lim_{T \rightarrow \infty} \frac{\sum_t R(G^v, t)}{\sum_t Cost(G^v, t)}. \quad (9)$$

In addition, the embedding delay $Delay(G^v, t)$ of the VNR G^v is defined by the following equation.

$$Delay(G^v, t) = \sum_{(n_i^s, n_j^s) \in L^E} D_{(n_i^s, n_j^s)}, \quad (10)$$

where L^E denotes the set of physical links embedded by the current VNR and $D_{(n_i^s, n_j^s)}$ denotes the delay of link (n_i^s, n_j^s) .

IV. EMBEDDING ALGORITHM

We customised a policy network to act as a reinforcement learning agent. It extracts the state of the network and derives node mapping schemes. The RL-based multi-domain virtual network embedding process is mainly plotted into two stages: node embedding and link embedding. In this section, we will introduce the implementation details of the multi-domain virtual network mapping algorithm that optimizes QoS.

A. K-Means-Based VNR's QoS Requirement Classification Model

Each VNR has a particular requirement for a certain QoS attribute. Before embedding the virtual network, we predict the QoS demand categories of VNRs based on the K-means algorithm. This allows our algorithm to adaptively and efficiently satisfy the differentiated QoS requirements of users. K-means algorithm is a common unsupervised clustering algorithm, which

is characterized by simple principle and high computational efficiency. The key to predicting the QoS requirement category of VNR using the K-means algorithm is the selection of the number of clusters k [26]. In virtual network embedding problems, common QoS requirements can be divided into requirements for computing resources, bandwidth, and delay. Therefore, the value of k is set to 3. The prediction result of K-means determines which reward function the agent uses. This allows the agent to make an embedding scheme that is more suitable for the current QoS requirements and does not lead to duplication of embedding schemes for VNRs. We use (11) as the clustering criterion to cluster the VNRs [27], [28]. Assuming that VNRs are divided into (C_1, C_2, C_3) , the goal is to minimize (12), where μ_i is the centroid of each category. After establishing a QoS category prediction model based on the K-means algorithm, in RL we select different reward signals according to the QoS requirement category of the VNR. The specific process of using K-means algorithm to classify VNR QoS requirements is as follows:

- 1) Clean the sample data of VNR to obtain data for experiment.
- 2) Determine the number of clusters K of the K-means algorithm, and then divide the sample data to obtain a K-means-based VNR's QoS requirement classification model.
- 3) Train and test the model, compare and analyze the difference between the predicted result and the actual value, and the hit rate of the K-means-based VNR's QoS requirement classification model is calculated.

$$d(x_i, x_j) = \sqrt{(x_i - x_j)^T (x_i - x_j)}. \quad (11)$$

$$E = \sum_{i=1}^3 \sum_{x \in C_i} \|x - \mu_i\|_2^2. \quad (12)$$

B. Feature Extraction

The key to using RL to evaluate the probability of a node being embedded lies in a precise understanding of the SAGOI-Net, which is conducive to the agent training in the SAGOI-Net as realistic as possible. Therefore, we abstract the following features for each substrate node and use them as the training environment of the agent [29], [30].

- 1) $CPU(n^s)$: The computing resources of physical node n^s determine its availability, and nodes with high computing power can meet the QoS requirements of different VNRs.
- 2) $DEG(n^s)$: In the substrate network topology, the degree of node n^s indicates the number of links connected to node n^s . The higher the degree of a node, the more likely it is to find a path to other physical nodes.
- 3) $SUM(n^s)_{BW}$: Each node has a set of links connected to it, $SUM(n^s)_{BW}$ is the sum of bandwidth resources of these links. The larger the value, the more bandwidth resources can be selected when the virtual node is mapped to n^s , and more links can be embedded.

$$SUM(n^s)_{BW} = \sum_{l^s \in L(n^s) \cup E(n^s)} BW(l^s), \quad (13)$$

where $L(n^s)$ represents the intra-domain link connected to node n^s , and $E(n^s)$ represents the inter-domain link connected to node n^s .

- 4) $SUM(n^s)_D$: In the same way, $SUM(n^s)_D$ is the sum of the maximum delay values of these links. The smaller the value is, the less the delay interference will be when the virtual node is mapped to n^s .

$$SUM(n^s)_D = \sum_{l^s \in L(n^s) \cup E(n^s)} D(l^s). \quad (14)$$

- 5) $AVG(n^s)_{Dst}$: After the node n^s is embedded, to cut down the bandwidth resource consumption of the VNR and reduce the impact of the delay on the VNR, it is also necessary to consider the positions of other non-embedded virtual nodes in the same VNR. We use (15) to calculate the average distance from this node to other non-embedded nodes.

$$AVG(n^s)_{Dst} = \frac{\sum_{n_i^s \in N^S} Dst(n^s, n_i^s)}{|N^S| + 1}. \quad (15)$$

In fact, the features that can be extracted in the SAGOI-Net are far more than the above five points. The more features that are extracted, the more realistic the training environment of the agent is, but the complexity of the algorithm increases accordingly. Therefore, after comprehensively considering the actual situation of SAGOI-Net, it is most appropriate to extract the above five features. After extracting the characteristics of the physical nodes from the three network segments of SAGOI-Net, we normalize them to feature vectors, as shown in (16).

$$v_k = (CPU(n_k^s), DEG(n_k^s), SUM(n_k^s)_{BW}, SUM(n_k^s)_D, AVG(n_k^s)_{Dst})^T, \quad k = Space, Air, Grond. \quad n_k^s \in N^S. \quad (16)$$

We combine the above vectors to form a feature matrix M_f . Each row of the matrix is a feature vector of a physical node. As SAGOI-Net's resources change, the feature matrix is constantly updated.

$$M_f = (v_1, v_2, \dots, v_{|N^S|})^T. \quad (17)$$

The specific expression of M_f is as follows:

C. Policy Network

The key to the realization of the multi-domain virtual network embedding algorithm based on QoS optimization is the design of the agent. We define the feature matrix of the network environment as the state space of the agent, as shown in (18) shown at the bottom of the next page. Then, the agent takes actions based on the states, i.e., possible node embedding schemes. Finally, different reward functions are chosen to measure the merit of the actions taken by the agent according to the QoS categories.

As shown in Fig. 3, we design a four-layer policy network as an agent, containing an input layer, a convolutional layer, a soft-max layer, and an output layer. We select a physical node for each virtual node to map through the policy network. In the input layer, we use the feature matrix extracted from SAGOI-Net

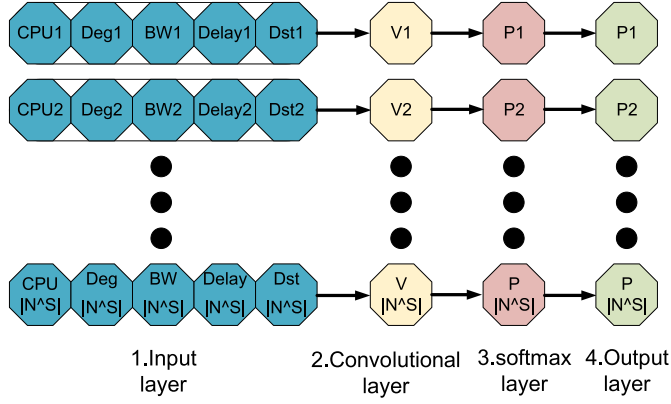


Fig. 3. Policy network.

as the input of the agent. In the convolution layer, we evaluate the resources of each physical node according to the extracted feature matrix, and generate a vector of available resources for each node through convolution operation. The convolution operation is as follows:

$$h_i^c = w \cdot v_i + b, \quad (19)$$

among them, $\cdot$ denotes the vector performing the dot product operation, h_i^c is the i -th output of the convolution layer, w is the weight vector of the convolution kernel, v_i is the i -th input of the convolutional layer, and b is the deviation. The convolution operation of the feature matrix by the above formula is actually the transformation calculation of the feature matrix and the convolution kernel.

In the softmax layer, through the softmax analyzer, we calculate the embedding probability of each physical node according to the available resource vector. The probability calculation formula is as follows:

$$P_i = \frac{e^{h_i^c}}{\sum_j e^{h_j^c}}. \quad (20)$$

The probability determines the possibility of the physical node being embedded by the virtual node. However, some physical nodes cannot load virtual nodes due to insufficient computing resources, so the probability of being embedded cannot be deduced. In order to solve this problem, we pre-selected the physical nodes to ensure that the nodes have sufficient computing resources.

It should be noted that since we pre-selected physical nodes in advance, some physical nodes with less computational resources were not used, which resulted in a certain waste of resources. However, these wastes are insignificant relative to the benefits of meeting users' high QoS requirements.

D. Training and Testing

We use historical network request data to construct a training set and a test set for training the agent to find better solutions and to evaluate the effectiveness of the algorithms, respectively. Firstly, we initialize the policy network parameters. For each arriving VNR, the agent will abstract a feature matrix from SAGOI-Net as the input. After calculating the embedding probability of each physical node according to (20), we cannot select physical nodes for embedding only in line with the probability, because the parameters of the agent are initialized randomly, which indicates that the probability is biased, and there may be other better solutions. Therefore, we generate samples from the set of physical nodes based on the probability distribution of all physical nodes, and select the physical nodes that are finally embedded. Then we use k-shortest path algorithm to complete the link embedding.

The main purpose of using the k-shortest path algorithm for link embedding is to improve the VNR acceptance rate. If the bandwidth of the substrate network link is insufficient, it can result in the underutilization of links with low bandwidth but low latency. This may fail to meet the user's demand for low latency and reduce the VNR acceptance rate.

When using the k-shortest path algorithm for link embedding, it is first necessary to select at most k shortest paths for the embedding node pair, and then embedding from small to large. If there is a path that meets the resource requirements of the virtual link, the embedding is successful and the current VNR is accepted. On the contrary, if these k paths do not meet the bandwidth and delay requirements of the virtual link, the embedding fails and the current VNR is rejected.

Throughout the training process, the agent depends on the reward signal to determine whether it is working correctly. The selection of the reward directly affects the entire training process. The agent depends on the result of the reward signal to determine whether the current behavior continues. Therefore, in the training process, to meet the QoS requirements of different smart applications, we use K-means to predict the QoS requirement category of each VNR, and select the appropriate reward signal in line with the QoS requirement category. This paper selects delay and revenue-cost ratio as reward signals. When the QoS requirement category of VNR is high bandwidth requirement or high computing resource requirement, RL uses the revenue-cost ratio as a reward signal. Otherwise, when the QoS requirement is a low latency requirement, the RL agent uses the ratio of bandwidth to latency as a reward signal. The larger the reward signal is, it means that the current node selection strategy of the agent leads to a greater revenue from virtual network embedding and the agent is working properly. On the contrary, the agent

$$\begin{bmatrix} CPU(n_1^s) & DEG(n_1^s) & SUM(n_1^s)_{BW} & SUM(n_1^s)_D & AVG(n_1^s)_{Dst} \\ CPU(n_2^s) & DEG(n_2^s) & SUM(n_2^s)_{BW} & SUM(n_2^s)_D & AVG(n_2^s)_{Dst} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ CPU(n_k^s) & DEG(n_k^s) & SUM(n_k^s)_{BW} & SUM(n_k^s)_D & AVG(n_k^s)_{Dst} \end{bmatrix}. \quad (18)$$

needs to adjust its behavior. The specific calculation methods of the two reward signals are as follows:

$$r_{r/c} = \frac{R(G^V, t)}{Cost(G^V, t)}. \quad (21)$$

$$r_{delay} = \frac{\sum_{(n_i^v, n_j^v) \in L^V} BW_{(n_i^v, n_j^v)}}{Delay(G^V, t)} \beta, \quad (22)$$

where β is an adjustable proportional value used to equalize the reward value size. In our work, we set it to 0.1.

Different from supervised learning, in RL, in order to calculate the loss value of the training process, this paper sets a label for each virtual node n^v in VNRs to represent the embedded physical node. And use 0, 1 to indicate whether the physical node is embedded. The embedding results of all physical nodes form a vector. For example, we set the label of virtual node n_k^v to k, which means that the k-th physical node is embedded, and the result vector can be expressed as follows:

$$v_{result} = (0_1, 0_2, \dots, 1_k, \dots, 0_n)^T. \quad (23)$$

Then we calculate the cross entropy loss based on the target result and label:

$$l(v_{result}, p) = - \sum_k v_{result}^k \log(p_k). \quad (24)$$

Finally, we use the following formula to calculate the gradient of the agent parameters:

$$g_l = g_f \cdot \alpha \cdot rw, \quad (25)$$

where g_f is the gradient size, α is the learning rate, and rw is the reward signal. In order to ensure the stability and efficiency of model training, we manually adjust the learning rate to control the gradient size and calculation speed. The training process of multi-domain virtual network embedding algorithm based on QoS optimization is shown in Algorithm 1.

In the test phase, we directly complete the embedding of the virtual node in line with the embedding probability of the SAGOI-Net node generated by the agent. Then, the k-shortest path algorithm is used to sequentially carry out the embedding of intra-domain links and inter-domain links. The code of the test process is indicated in Algorithm 2.

E. Complexity analytics

In this section, we analyze the time complexity of our proposed algorithm. First, we assume that there are n nodes in the physical network. We extract 5 feature attributes of the nodes. In the node embedding phase, the complexity of extracting the feature matrix from the physical network is $O(5n)$. Then, the time complexity of solving all feature vectors is $O(n^2)$. In addition, the feature matrix needs to be updated once every time a node is successfully embedded. Therefore, if all m nodes in the VNR are embedded successfully, the complexity of updating the feature matrix is $O(mn^2)$. In the link embedding phase, we implement the k-shortest path algorithm based on the breadth-first search algorithm, and the time complexity can be calculated as $O(kn)$. The final time complexity $T(n)$ can be calculated using the

Algorithm 1: Training.

```

1: Input: Parameters of policy network;
2: Output: The embedded probability of each physical
   node in SAGOI-Net;
3: Initialize all agent's parameters;
4: while iteration  $\leq$  Epoch do
5:   for VNR  $\in$  trainingSet do
6:     K  $\leftarrow$  means(VNR);
       %Predict the QoS requirement category of VNR
7:     reward  $\leftarrow$  selectReward();
8:     Mf  $\leftarrow$  getFeatureMatrix();
9:     prodis  $\leftarrow$  agent.getOutput(Mf);
       %Get the probability distribution of each node in
       SAGOI-Net
10:    Candidatens  $\leftarrow$  sample(prodis);
11:    getGradient(Candidatens);
12:    if isMapped( $\forall n^v \in N^V$ ) then
13:      k  $\leftarrow$  shortestpathLinkMap(req);
14:    end if
15:    if isMapped( $\forall n^v \in N^V, \forall l^v \in N^V$ ) then
16:      Calculate reward;
17:    else
18:      Clear gradients;
19:    end if
20:  end for
21:  iteration  $++$ ;
22: end while

```

Algorithm 2: Test.

```

1: Input: testingSet;
2: Output: Three performance indicators;
3: Initialize all agent's parameters;
4: for VNR  $\in$  testSet do
5:   K  $\leftarrow$  means(VNR);
6:   for  $n^v \in N^V$  do
7:     Mf  $\leftarrow$  getFeatureMatrix(Candidatens);
8:     getProbability();
9:   end for
10:  k  $\leftarrow$  shortestpathLinkMap(req);
11:  if isMapped( $\forall n^v \in N^V, \forall l^v \in N^V$ ) then
12:    return success;
13:  end if
14: end for

```

following equation.

$$T(n) = mn^2 + n^2 + kn + 5n. \quad (26)$$

V. EXPERIMENTAL DESIGN AND RESULT ANALYSIS

A. Parameters and Environment Settings

To compare the performance of our proposed RL and K-means-assisted QoS-aware virtual network embedding algorithm with other RL-based virtual network embedding algorithms, we conduct simulation experiments in a simulated

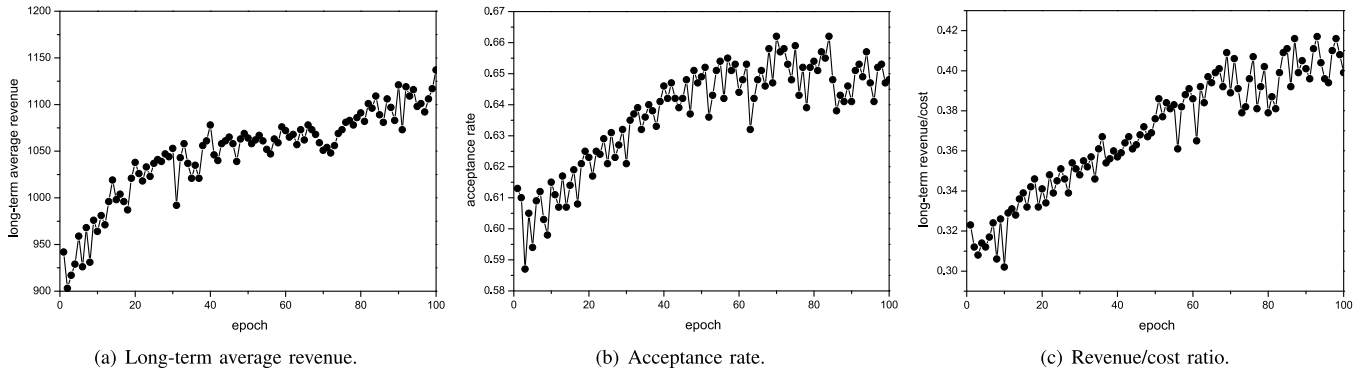


Fig. 4. Agent performance on training set.

network topology environment. At the same time, in order to simulate the SAGOI-Net as realistic as possible, we construct a three-layer network topology consisting of 100 physical nodes and about 500 physical links, including 20 space network nodes, 30 air network nodes, and 50 ground network nodes. In addition, in order to reduce the complexity of the solution, we use georgia tech internet work topology models (GTITM) for auxiliary solution. The transit-stub model provided by GTITM can well regenerate the hierarchical structure network. Therefore, it is very suitable for topological production of SAGOI-Net. According to the characteristics of different network segments, in the space network and the air network, the computing resources of each node are evenly distributed in [20,40], the bandwidth resources of each link in the domain are uniformly distributed in [50,100], and the delay value is evenly distributed in [1,10]. In the ground network, the computing resources of each node obey the uniform distribution of [50,100], the bandwidth resources of each link in the domain are randomly distributed between [50,100], and the delay value is randomly distributed between [1-10]. In addition, the bandwidth resources of the inter-domain link constituting the inter-domain communication are uniformly distributed in [50-100], and the delay value is randomly distributed between [1-15].

To make the agent work more accurately and efficiently, we generated 2000 VNRs, of which 1000 were used as the training set and 1000 were used as the test set. The number of nodes in each VNR is randomly distributed between [2,10], the computing resource requirements of virtual nodes are uniformly distributed in [1,20], the bandwidth resource demand of the virtual link is uniformly distributed in [5-15], and the delay demand is uniformly distributed in [1-10]. All the above parameter settings are shown in Table I.

B. Training Results and Analysis

Since the agent is not familiar with the SAGOI-Net, it will take some time to stabilize, especially the multi-domain virtual network embedding is NP-hard. We operate the policy network agent on the training set for 100 epochs and observe its performance. Fig. 4 indicates the change of agent in the training process from three aspects: long-term average revenue, virtual request acceptance rate and revenue cost ratio. As revealed

TABLE I
SIMULATION EXPERIMENT PARAMETER SETTING

Parameter name	Value
Number of physical nodes	100
Number of physical links	500
Space network nodes	20
Air network nodes	30
Ground network nodes	50
Computing resources of Space network nodes	[20,40]
Bandwidth resources of Space network links	[50,100]
Delay values of Space network links	[1,10]
Computing resources of Air network nodes	[20,40]
Bandwidth resources of Air network links	[50,100]
Delay values of Air network links	[1,10]
Computing resources of Ground network nodes	[50,100]
Bandwidth resources of Ground network links	[50,100]
Delay values of Ground network links	[1,10]
Number of virtual nodes	[2,10]
Computing resource required by virtual virtual nodes	[1, 20]
Bandwidth resources required by virtual links	[5, 15]
Delay requirements of virtual links	[1,10]
Bandwidth resources of inter-domain links	[50,100]
Delay values of inter-domain links	[1, 15]

in Fig. 4, because the parameters of the agent have just been initialized, the agent is not familiar with the SAGOI-Net, so the agent does not perform well at the beginning, and cannot give a good multi domain virtual network embedding scheme. As training proceeds, the agent gradually becomes familiar with the network environment of SAGOI-Net and can explore better multi-domain virtual network embedding decisions based on the reward signals. At the later stage of the training process, due to the limited ability of agent to solve multi domain virtual network embedding decision, the three indexes of agent tend to be stable. From the training results, it can be seen that using RL to solve multi domain virtual network embedding is effective.

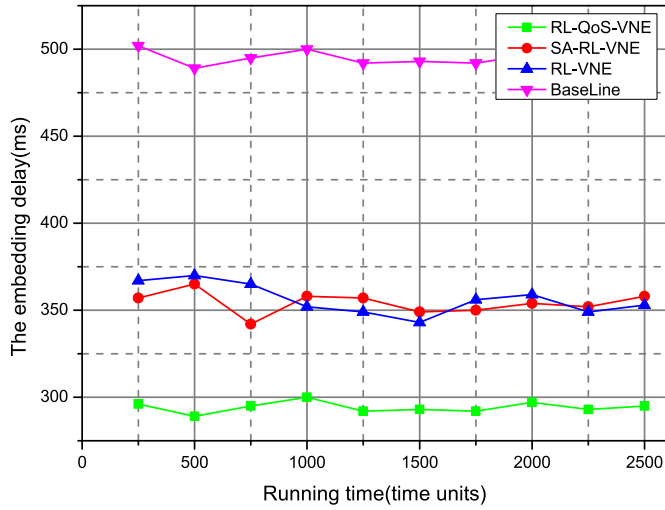


Fig. 5. The embedding delay comparison of four algorithms.

C. Test Results and Analysis

In the test phase, in order to verify the performance of the reinforcement learning-assisted QoS-aware virtual network embedding algorithm (RL-QoS-VNE) proposed in this paper, we compared it with the SA-RL-VNE proposed in [31], the RL-VNE proposed in [32], and the BaseLine algorithm proposed in [33]. In addition, in order to verify that the agent can still make good decisions when the QoS requirements change, this paper set VNRs to two categories: low delay requirements, high bandwidth, and high computing resources, and discussed the performance of the four algorithms in dealing with different QoS requirements. In the test phase, we divided the experiment into two parts.

When VNRs require low latency, we designed two experiments to compare four algorithms from the perspective of latency performance and VNR acceptance rate.

Experiment 1: Comparison of embedding delay of four algorithms

When the delay requirement of VNR is selected as a factor, Fig. 5 indicates the performance of the four algorithms at different time nodes. Since the RL-QoS-VNE algorithm proposed in this paper selects the total embedding delay as the reward function during the training process, the agent always aims to find the embedding strategy with the lowest delay. Although SA-RL-VNE and RL-VNE also use the RL algorithm to solve the virtual network embedding problem, they mainly consider the safety and revenue problems in the virtual network embedding process, and do not pay attention to the delay problem in the virtual network embedding process. Therefore, the RL-QoS-VNE algorithm is superior to other algorithms in terms of delay. In line with the analysis of specific experimental data, the embedding delay of RL-QoS-VNE is 17.14% lower than SA-RL-VNE, 17.29% lower than RL-VNE, and 40.81% lower than BaseLine.

Experiment 2: VNR acceptance rate of four algorithms(Take time delay as a factor)

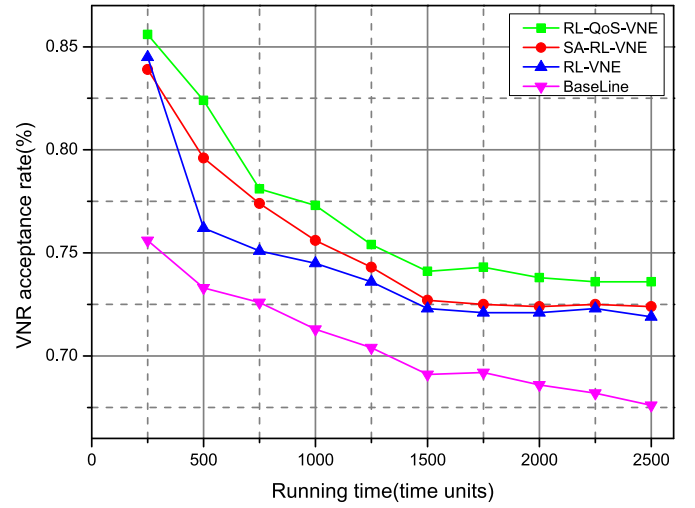


Fig. 6. The VNR acceptance rate of four algorithms(Take time delay as a factor).

Experiment 2 compares the VNR acceptance rates of the four algorithms on the basis of Experiment 1, and the experimental results are revealed in Fig. 6. Obviously, the VNR acceptance rates of the four algorithms decrease over time. Because RL-QoS-VNE considers the delay when solving virtual network embedding, when VNRs are low-latency requirements, SAGOI-Net can load more low-latency VNR compared to the other three algorithms, and the VNR acceptance rate is relatively higher. In line with the analysis of specific experimental data, the acceptance rate of RL-QoS-VNE is 2.7% higher than SA-RL-VNE, 2.6% higher than RL-VNE, and 8.5% higher than BaseLine.

When VNRs require high bandwidth or high computing resources, we designed three experiments to compare four algorithms from the perspective of long-term average revenue, revenue-cost ratio, and VNR acceptance rate.

Experiment 3: Long-term average revenue of the four algorithms

When the requirements for VNRs are high bandwidth or high computing resources, we set the reward function as the revenue-cost ratio. The result of Experiment 3 is revealed in Fig. 7. Obviously, due to the abundant resources available to SAGOI-Net in the early stage of virtual network embedding, the long-term average revenue of the four algorithms in the first 1000 time units (ms) are all higher. However, as the number of VNR increases, SAGOI-Net's resources decrease and cannot meet the subsequent VNR, so the revenue decreases rapidly. Since RL-QoS-VNE uses the K-means algorithm to predict the QoS requirements of the VNR, and uses the RL algorithm to rationally arrange the resources of SAGOI-Net, it can achieve better revenue than the other three algorithms.

Experiment 4: The revenue-cost ratio of the four algorithms

Fig. 8 reveals the results of Experiment 4, which denotes the revenue-cost ratio of the four algorithms. It can be seen from Fig. 8 that the revenue-cost ratio of the four algorithms does not decrease with the reduction of SAGOI-Net's available resources. This is mainly because the revenue-cost ratio is not

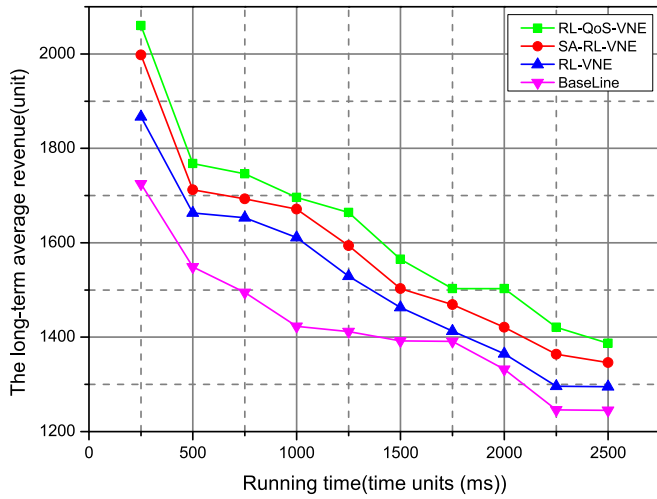


Fig. 7. Long-term average revenue of the four algorithms.

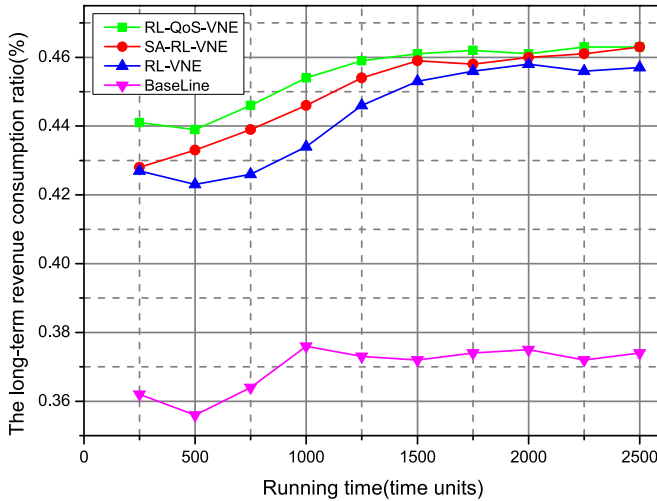


Fig. 8. The revenue-cost ratio of the four algorithms.

affected by changes in the underlying resources, and it mainly reflects the efficiency of the algorithm. Compared with the other three algorithms, because RL-QoS-VNE considers the various characteristics of SAGOI-Net when using RL to solve the virtual network embedding problem, and is targeted to the QoS of VNR, it can provide a better virtual network embedding strategy, and the revenue-cost ratio is also higher. According to the analysis of specific data, the revenue-cost ratio of RL-QoS-VNE is 1.1% higher than SA-RL-VNE, 1.4% higher than RL, and 23.7% higher than BaseLine.

experiment 5: VNR acceptance rate of four algorithms(Take bandwidth and computing resources as factors)

Similar to Experiment 2, Experiment 5 mainly discusses the VNR acceptance rate of the four algorithms when facing VNR with high bandwidth or high computing resources. The experimental results are shown in Fig. 9. Since SAGOI-Net in the early stage of virtual network embedding has more available resources and can load more VNR, the VNR acceptance rate is high. As

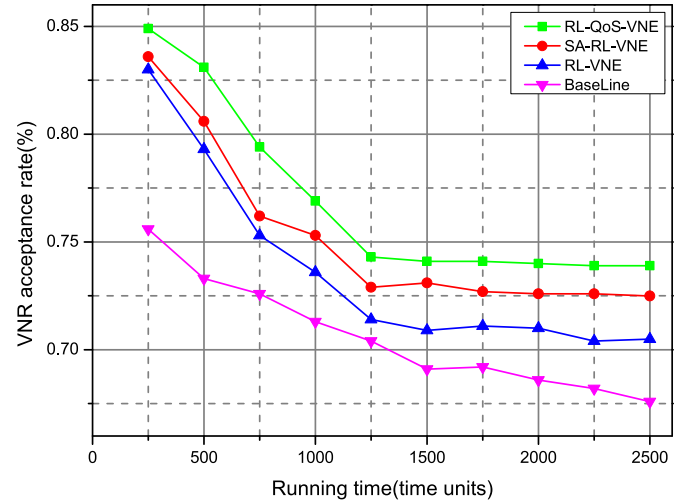


Fig. 9. VNR acceptance rate of four algorithms(Take bandwidth and computing resources as factors).

time goes by, the VNR acceptance rate gradually decreases. In line with specific experimental data, the VNR acceptance rate of RL-QoS-VNE is 1.3% higher than SA-RL-VNE, 2.7% higher than RL, and 8.5% higher than BaseLine.

Based on the above experimental results, we can conclude that the RL-QoS-VNE algorithm can effectively perceive the QoS requirements of VNR and can give a good virtual network embedding strategy, which is of practical significance. In addition, it is superior to SA-RL-VNE, RL-VNE and BaseLine in terms of embedding delay, long-term average revenue, and VNR acceptance rate, which shows that it has excellent performance.

VI. CONCLUSION

SAGOI-Net can take advantage of high flexibility, high reliability and high coverage by integrating space network, air network, and ground network. However, the seamless integration of the three heterogeneous network segments and the effective utilization of heterogeneous resources are still a problem. Based on SDN and network virtualization, we propose an RL-based QoS-aware virtual network embedding algorithm. This algorithm can effectively classify VNR before virtual network embedding, and then adjust different reward functions according to the type of VNR to motivate the model to make decisions that are more in line with user QoS requirements. The experimental results show that the RL-QoS-VNE proposed in this paper has certain flexibility in processing time-varying network virtual network embedding, and can effectively improve the resource utilization efficiency of SAGOI-Net, and meet the differentiated QoS requirements of users.

However, SAGOI-Net's attributes and topological structure are much more complicated in actual situations. Therefore, in the future work, we will extract more reasonable features of SAGOI-Net to make the training environment more realistic. In addition, we will study how to automatically sense the QoS requirements of VNR.

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